

Jonathan Vogel's discussion of

Quantifying the Sensitivity of Quantitative Trade Models

Chenguiti, Donaldson, and Wiles (CDW)

Princeton IES 2026 Summer Trade Workshop

Revisiting ACR

- ▶ ACR had four messages:

1. GFT is the area under the import demand curve
2. To trace that curve, require an elasticity function
3. In a wide class of models, that function is constant
4. In that class, $\exists$ a common, most-data-disciplined approach to calculate GFT

- ▶ CDW: matching ACR moments does not imply discipline outside this class

My three points

1. Can conclude negatively on robustness of ACR approach w/out computation
 - ▶ matching λ_0 and the local elasticity θ_0 allows GFT in $(0, \infty]$
2. Yet wide GFT range is not the same as ACR approach being a bad predictor
 - ▶ at worst-case bound, prediction may be almost perfect for almost every country
3. Can CDW diagnose why ACR approach fails if/when it does so?
 - ▶ is it variable elasticity, country heterogeneity, or the wrong elasticity moment?
 - ▶ a more constructive paper if CDW could help identify key failure

Gains from trade review (compensating variation moving from trade to autarky)

Ricardian model equivalent to a pure exchange model (Wilson, 1980)

Home endowment Y ; normalize price to one

Home initial utility u_0

p price of Hicks composite across ROW

Define $\mathcal{E}(p) \equiv e(1, p, u_0)$

Money metric gains from trade (in levels)

$$\begin{aligned} GFT(p_0) &= \mathcal{E}(\infty) - \mathcal{E}(p_0) \\ &= \mathcal{E}(\infty) - Y \end{aligned}$$

log gains

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log gains

For any p_1

$$\mathcal{E}(p_1) - \mathcal{E}(p_0) = \int_{p_0}^{p_1} \mathcal{E}'(p) dp = \int_{p_0}^{p_1} m(p) dp$$

Used Shephard's lemma: $m(p) \equiv \mathcal{E}'(p)$
is compensated import demand

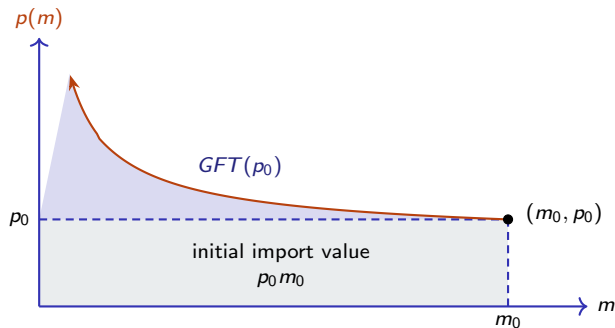
Taking limit as $p_1 \rightarrow \infty$ gives

$$GFT(p_0) = \int_{p_0}^{\infty} m(p) dp$$

Let $p(m)$ be inverse compensated import demand

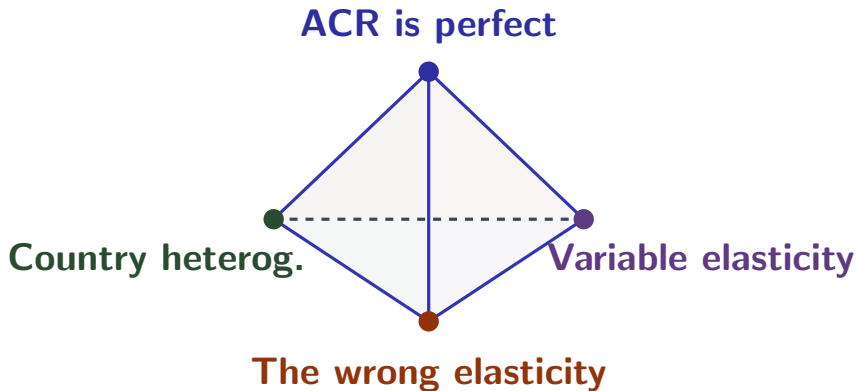
$$GFT(p_0) = \int_0^{m_0} [p(m) - p_0] dm$$

GFT in a picture



- ▶ GFT is area under inverse compensated import demand above price p_0
- ▶ Sufficient statistic: trade elasticity $\theta(p)$ for all $p \geq p_0$

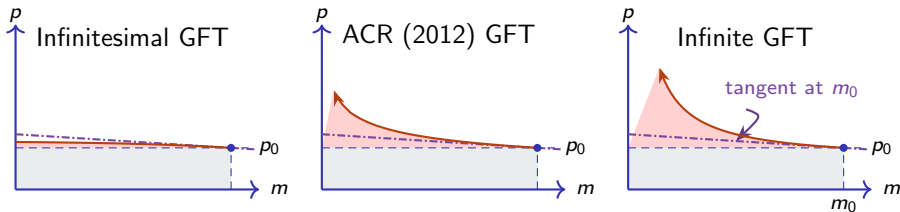
When is the ACR approach imperfect?



- ▶ GFT derivation/figure: one country, variable elasticity, the right elasticity
- ▶ Each of my points isolates one issue

Point 1: A pen-and-paper negative result

- Suppose at date 0 we target θ_0 (elasticity of compensated import demand in a country) and λ_0 (that country's domestic expenditure share)



GFT range (conditional on matching θ_0 and λ_0) is $(0, \infty]$

(ACD show theoretically possible to trace curve)

Conclusion: if objective is to show that ACR approach can fail, GFT range is $(0, \infty]$ without solving robust-control problem

Point 2: Wide bounds need not imply poor prediction

A simplified example in their spirit

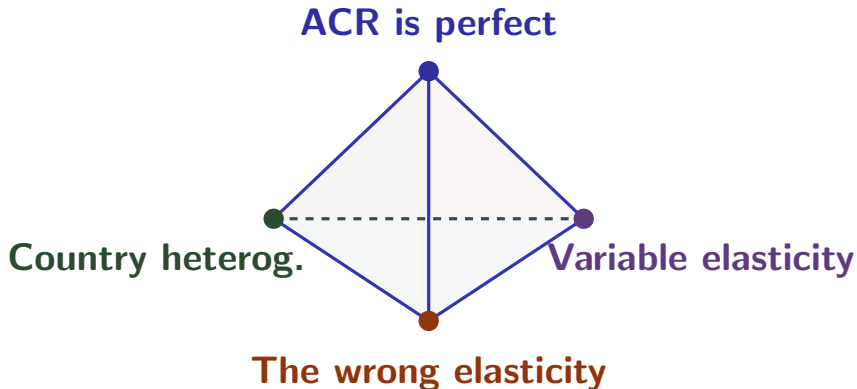
- ▶ Calibrate EK with many countries, drawing productivities
- ▶ Now make France Armington: It produces only wine and no one else can...
- ▶ Estimate of moment in $M_{\text{Model}} G_{\text{Generated}} D_{\text{Data}}$: barely below Fréchet parameter
- ▶ France's GFT is ACR formula $w / (\sigma - 1)$. All others almost exactly ACR with θ

Conclusion: achieve robust-control upper bound, but EK/ACR prediction is almost perfect almost everywhere

Discuss what we learn from this type of robustness exercise

Point 3: Which restriction of ACR is failing?

- ▶ The first two examples make two separate points:
 - ▶ wide bounds are easy to generate (variable elasticity)
 - ▶ wide bounds do not necessarily imply poor prediction (heterogeneity)
- ▶ Can we diagnose which economic restriction is failing?



Point 3: ACR can fail for a # of reasons (wrong elasticity)

- ▶ Suppose θ constant, common across countries: ACR formula applies
all is for the best in the best of all possible worlds [Dr. Pangloss]

Point 3: ACR can fail for a # of reasons (wrong elasticity)

- ▶ Suppose θ constant, common across countries: ACR formula applies
all is for the best in the best of all possible worlds [Dr. Pangloss]
- ▶ ACR model class is a tiny sliver of the model space where this formula applies
- ▶ It is special: Only here is θ identified from variation btw foreign sources
- ▶ Generically, our estimate is unrelated to θ when the formula holds
The standard approach raises some obvious concerns. One unfortunate feature of the empirical estimates ... is that they come mostly from variation in trade costs and trade flows across foreign countries. Indeed, the previous regressions are often run excluding domestic flows. [Costinot Rodriguez-Clare, JEP (2018), pgs 14-15]
- ▶ Away from Fréchet-Pareto, this moment is not the right elasticity
Re-stating conclusion: If we know this is the reason for failure (or not), focus on trying to estimate elasticity of import share

Summary

- ▶ It's an interesting technique to bring to trade
- ▶ Is a wide GFT range indicative that ACR prediction is unlikely to be good?
- ▶ I am receptive to CDW's skepticism, based on pen-and-paper examples
- ▶ Can they suggest where to focus? That would be a very nice contribution

Appendix

Gains from trade review (compensating variation moving from autarky to trade)

Ricardian model equivalent to a pure exchange model (Wilson, 1980)

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Define $\mathcal{E}(p) \equiv e(1, p, u_0)$

Gains from trade

$$GFT(p_0) \equiv \log \frac{\mathcal{E}(\infty)}{\mathcal{E}(p_0)} = \log \frac{\mathcal{E}(\infty)}{Y}$$

For any p_1 ,

$$\log \frac{\mathcal{E}(p_1)}{\mathcal{E}(p_0)} = \int_{\log p_0}^{\log p_1} \frac{d \log \mathcal{E}(p)}{d \log p} d \log p = \int_{\log p_0}^{\log p_1} \lambda_F(p) d \log p$$

Used Shephard's lemma:

$$\frac{d \log \mathcal{E}(p)}{d \log p} = \frac{p \mathcal{E}'(p)}{\mathcal{E}(p)} = \frac{p m(p)}{\mathcal{E}(p)} = \lambda_F(p).$$

Taking $p_1 \rightarrow \infty$ gives

$$GFT(p_0) = \int_{\log p_0}^{\infty} \lambda_F(p) d \log p.$$

Sufficient conditions for ACR upper bound

[back to CDW slide](#)

Let $m^{ACR}(p)$ be compensated import demand curve with constant trade elasticity θ , and suppose two economies start from the same (p_0, m_0, λ_0) . Leverage this identity

[derivation](#)

$$-\frac{d \log m(p)}{d \log p} = \lambda(p)[1 + \theta(p)]$$

If $\theta(p) \geq \theta$ for all $p \geq p_0$, then

$$\frac{d}{d \log p} \log \frac{\lambda(p)}{1 - \lambda(p)} \equiv \theta(p) \geq \theta \equiv \frac{d}{d \log p} \log \frac{\lambda^{ACR}(p)}{1 - \lambda^{ACR}(p)}.$$

Since both economies start from the same λ_0 , for all $p \geq p_0$ we have

$$\lambda(p) \geq \lambda^{ACR}(p)$$

Therefore

$$-\frac{d \log m(p)}{d \log p} = \lambda(p)[1 + \theta(p)] \geq \lambda^{ACR}(p)[1 + \theta] = -\frac{d \log m^{ACR}(p)}{d \log p}$$

Since $m(p_0) = m^{ACR}(p_0) = m_0$, this implies that for all $p \geq p_0$

$$m(p) \leq m^{ACR}(p)$$

Hence

$$GFT(p_0) = \int_{p_0}^{\infty} m(p) dp \leq \int_{p_0}^{\infty} m^{ACR}(p) dp = GFT^{ACR}(p_0)$$

Derivation of $d \log m(p)/d \log p$

[back to sufficient condition](#)[back to GFT range](#)

- ▶ We have (1) $m(p)p = (1 - \lambda(p))\mathcal{E}(p)$ and (2) $d \log \mathcal{E}(p)/d \log p = 1 - \lambda(p)$
- ▶ Log differentiating (1) yields

$$\frac{d \log m(p)}{d \log p} = \frac{d \log(1 - \lambda(p))}{d \log p} + \frac{d \log \mathcal{E}(p)}{d \log p} - 1$$

- ▶ Substituting in (2) and in for $1 - \lambda(p)$ gives

$$\frac{d \log m(p)}{d \log p} = \frac{d \log(1 - \lambda(p))}{d \log p} + 1 - \lambda(p) - 1 = \frac{d \log(1 - \lambda(p))}{d \log p} - \lambda(p)$$

- ▶ Now use the definition of the trade elasticity $\theta(p) \equiv -\frac{d}{d \log p} \log \frac{1 - \lambda(p)}{\lambda(p)}$ to obtain

$$\theta(p) = -\frac{1}{\lambda(p)} \frac{d \log(1 - \lambda(p))}{d \log p}$$

- ▶ Combining yields

$$-\frac{d \log m(p)}{d \log p} = \lambda(p)[1 + \theta(p)]$$

CES benchmark

Suppose the trade elasticity is constant:

$$\theta(p) \equiv \theta > 0$$

With

$$\mathcal{R}_0 \equiv \frac{1 - \lambda_0}{\lambda_0} \quad \text{and} \quad t \equiv \ln(p/p_0)$$

the import-to-domestic expenditure ratio is

$$\mathcal{R}(t) = \mathcal{R}_0 e^{-\theta t}$$

Since

$$\frac{d \ln \mathcal{E}}{dt} = 1 - \lambda(t) = \frac{\mathcal{R}(t)}{1 + \mathcal{R}(t)}$$

we have

$$\ln \frac{\mathcal{E}(\infty)}{Y} = \int_0^\infty \frac{\mathcal{R}_0 e^{-\theta t}}{1 + \mathcal{R}_0 e^{-\theta t}} dt = \frac{1}{\theta} \ln(1 + \mathcal{R}_0) = -\frac{1}{\theta} \ln \lambda_0$$

Therefore

$$\frac{\mathcal{E}(\infty)}{Y} = \lambda_0^{-1/\theta} \quad \text{and} \quad GFT(p_0) = Y \left(\lambda_0^{-1/\theta} - 1 \right)$$